Functional summary

Honeybee head: proteins and RNA

MOFA | 20 matched samples | Proteomics + RNA | Workspace bee-head-example

Prepared from the completed browser run. 5 components; top 20 features per sign and view.

Feature alignment: matched_genes. Feature scaling: unit_variance. Center: on. Whole-view scaling: off. Maximum missingness: 30%. Seed: 557.

Component overview

Factor 1 | Metabolism and biosynthesis

Proteomics: 33.3% reconstruction | RNA: 2.4% reconstruction

0 term tests pass report-wide BH q < 0.05. Candidate.

Factor 2 | Gene-expression and biosynthetic regulation

Proteomics: 0.0% reconstruction | RNA: 33.9% reconstruction

29 term tests pass report-wide BH q < 0.05. BH-supported.

Factor 3 | Metabolism and biosynthesis

Proteomics: 0.1% reconstruction | RNA: 23.7% reconstruction

0 term tests pass report-wide BH q < 0.05. Candidate.

Factor 4 | Cellular homeostasis

Proteomics: 20.5% reconstruction | RNA: 0.1% reconstruction

0 term tests pass report-wide BH q < 0.05. Candidate.

Factor 5 | Metabolism and biosynthesis

Proteomics: 15.8% reconstruction | RNA: 1.9% reconstruction

0 term tests pass report-wide BH q < 0.05. Candidate.

Reconstruction basis: centered variation

Reading the report

BH-supported themes contain a term with report-wide BH q < 0.05. Candidate themes are ranked functional leads. Hypotheses are proposed experiments. Positive and negative weights describe opposite ends of a component; the sign can flip between fitted models.

COMPONENT PROFILE

Factor 1: Metabolism and biosynthesis

Functional themes

Metabolism and biosynthesis | Candidate

cellular catabolic process. Score 1.78; 3 related terms; lowest report BH q 0.25.

cellular catabolic process [GO:0044248]: Proteomics, negative; hits 3/7; report q=0.25.

catabolic process [GO:0009056]: Proteomics, negative; hits 3/7; report q=0.49.

IDs: LOC408358, LOC411557, LOC412843

Cellular homeostasis | Candidate

response to external stimulus. Score 1.43; 2 related terms; lowest report BH q 0.83.

response to external stimulus [GO:0009605]: RNA, negative; hits 3/10; report q=0.83.

response to stress [GO:0006950]: RNA, negative; hits 3/10; report q=0.83.

IDs: GeneID_412484, LOC409979, LOC412843, LOC552285

Cellular homeostasis | Candidate

cellular response to stimulus. Score 1.17; 2 related terms; lowest report BH q 1.0.

cellular response to stimulus [GO:0051716]: RNA, negative; hits 3/10; report q=1.0.

response to stimulus [GO:0050896]: RNA, negative; hits 4/10; report q=1.0.

IDs: GeneID_412484, LOC409979, LOC412843, LOC552285

Sample associations

group: Cranberry = 1: rho=-0.43, n=20; province: Quebec = 1: rho=0.20, n=20; rna_library_size: rho=-0.19, n=20

Hypotheses and proposed tests

Metabolism and biosynthesis

Factor 1 reconstructs more of the protein input (33.3%) than the RNA input (2.4%). Differences in translation or protein turnover could contribute to the cellular catabolic process signature.

Test: Measure the highlighted transcripts alongside new protein synthesis and protein half-lives in independently collected matched samples.

Study context: [1]

Cellular homeostasis

One hypothesis is that cellular maintenance or stimulus-response state differs across the samples, involving the highlighted response to external stimulus genes/features.

Test: Compare the highlighted proteins with tissue composition and a controlled stimulus-response assay.

Study context: [1]

Dscam follow-up

Dscam ranks among the strongest features for Factor 1. A follow-up could test whether its expression or splice forms track the measured sample differences.

Test: Measure Dscam isoforms by targeted RNA sequencing and its protein abundance in independent head samples; pair the measurements with an appropriate learning assay.

Related research: [4]

Leading measured features

Proteomics: LOC100576993 (0.895), LOC551123 (0.876), LOC100578084 (0.858), LOC410687 (0.851)

RNA: Dscam (0.504), LOC100577697 (0.280), Obp4 (0.278), LOC409962 (-0.271)

COMPONENT PROFILE

Factor 2: Gene-expression and biosynthetic regulation

Functional themes

Gene-expression and biosynthetic regulation | BH-supported

regulation of biosynthetic process. Score 4.51; 18 related terms; lowest report BH q 0.00018.

regulation of biosynthetic process [GO:0009889]: RNA, positive; hits 10/17; report q=0.00018.

regulation of cellular biosynthetic process [GO:0031326]: RNA, positive; hits 10/17; report q=0.00018.

IDs: LOC100576251, LOC100576993, LOC408877, LOC409456, LOC409928

Functional annotation profile | BH-supported

regulation of biological process. Score 4.28; 3 related terms; lowest report BH q 0.0071.

regulation of biological process [GO:0050789]: RNA, positive; hits 12/17; report q=0.0071.

regulation of cellular process [GO:0050794]: RNA, positive; hits 12/17; report q=0.0071.

IDs: LOC100576251, LOC100576993, LOC408576, LOC408877, LOC409456

RNA processing and gene regulation | BH-supported

RNA metabolic process. Score 3.07; 18 related terms; lowest report BH q 0.011.

RNA metabolic process [GO:0016070]: RNA, positive; hits 7/17; report q=0.011.

regulation of RNA metabolic process [GO:0051252]: RNA, positive; hits 7/17; report q=0.011.

IDs: LOC100576993, LOC408877, LOC409456, LOC409928, LOC411557

Sample associations

year: rho=0.76, n=20; rna_TMM_factor: rho=0.38, n=20; province: Quebec = 1: rho=-0.38, n=20

Hypotheses and proposed tests

Gene-expression and biosynthetic regulation

Factor 2 reconstructs more of the RNA input (33.9%) than the protein input (0.0%). The regulation of biosynthetic process signature could be buffered between RNA expression and protein accumulation.

Test: Track the highlighted genes at RNA and protein levels across matched time points, and compare the timing and size of their responses.

Related research: [5]

Functional annotation profile

Factor 2 reconstructs more of the RNA input (33.9%) than the protein input (0.0%). The regulation of biological process signature could be buffered between RNA expression and protein accumulation.

Test: Track the highlighted genes at RNA and protein levels across matched time points, and compare the timing and size of their responses.

Study context: [1]

Leading measured features

Proteomics: LOC725380 (-0.003), Def1 (-0.003), LOC413117 (0.003), LOC113218789 (-0.003)

RNA: LOC102655980 (0.971), LOC408740 (0.958), LOC551123 (0.951), LOC724592 (0.949)

COMPONENT PROFILE

Factor 3: Metabolism and biosynthesis

Functional themes

Metabolism and biosynthesis | Candidate

carboxylic acid metabolic process. Score 2.17; 6 related terms; lowest report BH q 0.16.

carboxylic acid metabolic process [GO:0019752]: RNA, negative; hits 4/10; report q=0.16.

organic acid metabolic process [GO:0006082]: RNA, negative; hits 4/10; report q=0.16.

IDs: LOC411188, LOC412166, LOC412843, LOC552249, LOC725400

Cell communication and adhesion | Candidate

cell adhesion. Score 2.10; 1 related terms; lowest report BH q 0.34.

cell adhesion [GO:0007155]: RNA, positive; hits 3/8; report q=0.34.

IDs: LOC409722, LOC410824, LOC412968

Metabolism and biosynthesis | Candidate

cellular catabolic process. Score 1.61; 2 related terms; lowest report BH q 0.60.

cellular catabolic process [GO:0044248]: RNA, negative; hits 3/10; report q=0.60.

catabolic process [GO:0009056]: RNA, negative; hits 3/10; report q=0.83.

IDs: LOC408358, LOC412843, LOC725400

Sample associations

rna_TMM_factor: rho=0.45, n=20; year: rho=-0.29, n=20; province: Quebec = 1: rho=0.29, n=20

Hypotheses and proposed tests

Metabolism and biosynthesis

Factor 3 reconstructs more of the RNA input (23.7%) than the protein input (0.1%). The carboxylic acid metabolic process signature could be buffered between RNA expression and protein accumulation.

Test: Track the highlighted genes at RNA and protein levels across matched time points, and compare the timing and size of their responses.

Related research: [5]

Cell communication and adhesion

Factor 3 reconstructs more of the RNA input (23.7%) than the protein input (0.1%). The cell adhesion signature could be buffered between RNA expression and protein accumulation.

Test: Track the highlighted genes at RNA and protein levels across matched time points, and compare the timing and size of their responses.

Related research: [4]

Leading measured features

Proteomics: LOC551123 (0.016), LOC552401 (-0.009), CSP3 (0.009), Apd-2 (-0.007)

RNA: LOC551323 (1.013), LOC552154 (0.985), LOC409722 (0.984), LOC724382 (0.981)

COMPONENT PROFILE

Factor 4: Cellular homeostasis

Functional themes

Cellular homeostasis | Candidate

cellular homeostasis. Score 1.69; 8 related terms; lowest report BH q 0.34.

cellular homeostasis [GO:0019725]: Proteomics, positive; hits 3/8; report q=0.34.

chemical homeostasis [GO:0048878]: Proteomics, positive; hits 3/8; report q=0.34.

IDs: LOC408306, LOC410860, LOC412843

Functional annotation profile | Candidate

Phagocytosis - Apis mellifera (honey bee). Score 1.89; 1 related terms; lowest report BH q 0.51.

Phagocytosis - Apis mellifera (honey bee) [ame04145]: Proteomics, negative; hits 3/9; report q=0.51.

IDs: LOC113218576, LOC724899, LOC726721

RNA processing and gene regulation | Candidate

DNA-templated transcription. Score 1.57; 13 related terms; lowest report BH q 0.64.

DNA-templated transcription [GO:0006351]: Proteomics, negative; hits 4/13; report q=0.64.

RNA biosynthetic process [GO:0032774]: Proteomics, negative; hits 4/13; report q=0.64.

IDs: LOC100576993, LOC408877, LOC409928, LOC724592

Sample associations

group: Cranberry = 1: rho=0.76, n=20; province: Quebec = 1: rho=0.49, n=20; year: rho=-0.47, n=20

Hypotheses and proposed tests

Cellular homeostasis

Factor 4 reconstructs more of the protein input (20.5%) than the RNA input (0.1%). Differences in translation or protein turnover could contribute to the cellular homeostasis signature.

Test: Measure the highlighted transcripts alongside new protein synthesis and protein half-lives in independently collected matched samples.

Study context: [1]

Functional annotation profile

Factor 4 reconstructs more of the protein input (20.5%) than the RNA input (0.1%). Differences in translation or protein turnover could contribute to the phagocytosis - apis mellifera (honey bee) signature.

Test: Measure the highlighted transcripts alongside new protein synthesis and protein half-lives in independently collected matched samples.

Study context: [1]

Leading measured features

Proteomics: LOC552154 (1.021), C1q-VP (0.989), LOC724993 (0.904), Apd-2 (0.903)

RNA: LOC551323 (-0.034), LOC408877 (-0.026), LOC410824 (0.019), LOC550930 (-0.017)

COMPONENT PROFILE

Factor 5: Metabolism and biosynthesis

Functional themes

Metabolism and biosynthesis | Candidate

organic substance metabolic process. Score 1.40; 5 related terms; lowest report BH q 0.51.

organic substance metabolic process [GO:0071704]: RNA, positive; hits 11/13; report q=0.51.

metabolic process [GO:0008152]: RNA, positive; hits 11/13; report q=0.64.

IDs: GeneID_406131, GeneID_412484, LOC100576895, LOC100577739, LOC409456

Nutrient and small-molecule metabolism | Candidate

organonitrogen compound metabolic process. Score 1.34; 2 related terms; lowest report BH q 0.64.

organonitrogen compound metabolic process [GO:1901564]: RNA, positive; hits 5/13; report q=0.64.

organonitrogen compound biosynthetic process [GO:1901566]: RNA, positive; hits 3/13; report q=1.0.

IDs: GeneID_412484, LOC100576895, LOC409807, LOC551123, LOC724704

Transport and cell organization | Candidate

transport. Score 1.20; 3 related terms; lowest report BH q 0.96.

transport [GO:0006810]: Proteomics, negative; hits 3/9; report q=0.96.

establishment of localization in cell [GO:0051649]: Proteomics, negative; hits 2/9; report q=1.0.

IDs: LOC409722, LOC724386, LOC726721

Sample associations

year: rho=0.75, n=20; rna_TMM_factor: rho=0.43, n=20; province: Quebec = 1: rho=0.14, n=20

Hypotheses and proposed tests

Metabolism and biosynthesis

One hypothesis is that metabolic or biosynthetic state varies across the samples, involving the highlighted organic substance metabolic process genes/features.

Test: Measure the highlighted enzymes together with targeted metabolites in independent samples.

Related research: [5]

Nutrient and small-molecule metabolism

One hypothesis is that nutrient use or small-molecule metabolism varies across the samples, involving the highlighted organonitrogen compound metabolic process genes/features.

Test: Quantify the corresponding metabolites and enzyme activities in independently collected matched samples.

Related research: [5]

Dscam follow-up

Dscam ranks among the strongest features for Factor 5. A follow-up could test whether its expression or splice forms track the measured sample differences.

Test: Measure Dscam isoforms by targeted RNA sequencing and its protein abundance in independent head samples; pair the measurements with an appropriate learning assay.

Related research: [4]

Leading measured features

Proteomics: LOC552401 (0.841), Def1 (0.776), CSP3 (0.733), LOC724704 (0.685)

RNA: Dscam (-0.544), LOC724704 (0.159), LOC100577376 (-0.158), LOC113218789 (0.147)

Methods and reproducibility

Signed top-weight gene sets; one-sided hypergeometric over-representation. Within-query and report-wide Benjamini-Hochberg correction. Terms with at least two query hits are grouped using complete-linkage clustering at kappa >= 0.5, with at least three shared model genes. Cluster score is mean(-log10(raw term p)); it summarizes related terms and is not an additional p value. This is a DAVID-inspired local grouping, using ordinary hypergeometric p values rather than DAVID EASE scores.

Report-wide correction covers 1904 term tests across components, views, annotation catalogs and weight directions. Complete-linkage term clustering uses kappa >= 0.5, with at least three shared model genes. Positive/negative queries select up to 20 features each. Views contributing at least 1% reconstruction enter annotation queries.

Settings

transform: auto

normalization: none

max_missing: 0.3

model_features: 100

standardize: true

feature_scaling: unit_variance

mofa_center: true

mofa_scale_views: false

factors: 5

iterations: 300

seed: 557

challenge: none

mask_fraction: 0.1

method: mofa

feature_alignment: matched_genes

References

[1] Omics Insights Into the Effects of Highbush Blueberry and Cranberry Crop Agroecosystems on Honey Bee Health and Physiology. Proteomics, 2025.

https://doi.org/10.1002/pmic.70033

[2] DAVID: a web server for functional enrichment analysis and functional annotation of gene lists (2021 update). Nucleic Acids Research, 2022.

https://doi.org/10.1093/nar/gkac194

[3] MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. Genome Biology, 2020.

https://doi.org/10.1186/s13059-020-02015-1

[4] Memory consolidation in honey bees is enhanced by down-regulation of Down syndrome cell adhesion molecule and changes its alternative splicing. Frontiers in Molecular Neuroscience, 2024.

https://doi.org/10.3389/fnmol.2023.1322808

[5] Imidacloprid disrupts larval molting regulation and nutrient energy metabolism, causing developmental delay in honey bee Apis mellifera. eLife, 2024.

https://doi.org/10.7554/elife.88772

Abbreviations and explanations

Pareto - Pareto scaling

Divides each feature by the square root of its standard deviation; centering is controlled separately.

rho - Spearman rank correlation coefficient

Describes how two quantities vary together by rank.

kappa - Cohen kappa similarity

Measures agreement between term-gene membership profiles beyond chance.

n - Observed sample count

The number of paired observations used for a result.

UniProt - Universal Protein Resource

A curated resource of protein sequences and functional annotations.

RNA - Ribonucleic acid

Carries and regulates genetic messages; RNA abundance is one molecular layer.

DNA - Deoxyribonucleic acid

Stores the genetic sequence.

MOFA - Multi-Omics Factor Analysis

Finds latent factors that describe variation across several molecular layers.

MOFA2 - Multi-Omics Factor Analysis, version 2

The R software used to fit and inspect MOFA models.

PCA - Principal Component Analysis

Summarizes the main directions of variation in a matrix.

PC - Principal Component

One axis obtained from principal component analysis.

GO - Gene Ontology

A structured vocabulary describing gene functions and biological processes.

BP - Biological Process

The Gene Ontology branch describing biological activities.

GO_BP - Gene Ontology: Biological Process

The biological-process annotation catalog used here.

KEGG - Kyoto Encyclopedia of Genes and Genomes

A database linking genes and compounds to pathways.

ORA - Over-Representation Analysis

Tests whether an annotation occurs more often in a selected feature list than in its background.

DAVID - Database for Annotation, Visualization and Integrated Discovery

A resource for gene annotation, enrichment and grouping related functional terms.

EASE - Expression Analysis Systematic Explorer

The modified Fisher-test score used by DAVID.

BH - Benjamini-Hochberg procedure

Adjusts p values for multiple comparisons.

FDR - False Discovery Rate

The expected proportion of false discoveries among selected results under the testing procedure.

q - BH-adjusted p value in this lab

A p value adjusted with the Benjamini-Hochberg procedure.

p - p value

The probability of a result at least this extreme under the specified null model.

R2 - Coefficient of determination

Fraction of the model input reconstructed relative to a zero baseline; with centering on, it describes variation around the mean.

K - Number of latent factors

The chosen number of factors in the model.

SD - Standard Deviation

Measures the spread of a feature. Unit-variance scaling divides by SD; Pareto scaling divides by its square root.

QC - Quality Control

Checks measurement gaps, scales and sample patterns.

CPM - Counts Per Million

RNA counts divided by the effective library size, multiplied by one million.

TMM - Trimmed Mean of M-values

An RNA library-size normalization method that trims extreme log-ratios.

TPM - Transcripts Per Million

RNA abundance adjusted for transcript length and library composition.

TMT - Tandem Mass Tag

An isotope-labeling method for multiplexed protein quantification.

MS - Mass Spectrometry

Measures ions by mass-to-charge ratio.

LC - Liquid Chromatography

Separates compounds before measurement.

LC-MS - Liquid Chromatography-Mass Spectrometry

Combines compound separation with mass-spectrometric detection.

Abbreviations, continued

MS/MS - Tandem Mass Spectrometry

Fragments ions to help identify molecules.

m/z - Mass-to-charge ratio

The mass of an ion divided by its charge.

RT - Retention Time

The time at which a compound elutes from a chromatography column.

HBB - Highbush Blueberry

One crop environment in the honeybee study.

CRA - Cranberry

The other crop environment in the honeybee study.

CCLE - Cancer Cell Line Encyclopedia

A collection of molecular measurements from cancer cell-line models.

T2D - Type 2 Diabetes

The disease group in the human-islet dataset.

RRID - Research Resource Identifier

The source identifier used to match the islet donors.

ID - Identifier

A stable label used to match samples or molecular features.

LOC - Locus identifier prefix

An NCBI label for a gene locus; its number is an identifier rather than a functional name.

GeneID - NCBI Gene Identifier

A database identifier for a gene.

NCBI - National Center for Biotechnology Information

Maintains Gene, PubMed and other biological databases.

WGCNA - Weighted Gene Co-expression Network Analysis

Groups genes with correlated expression patterns.

HTSeq - High-Throughput Sequencing analysis framework

Software used to count RNA-sequencing reads.

RNA-seq - RNA Sequencing

Measures RNA by sequencing RNA-derived molecules.

ATP - Adenosine Triphosphate

A molecule used for cellular energy transfer.

ROS - Reactive Oxygen Species

Reactive oxygen-containing molecules involved in signaling and oxidative stress.

TCA - Tricarboxylic Acid cycle

A central metabolic pathway that oxidizes carbon fuels.

Dscam - Down syndrome cell adhesion molecule

A cell-adhesion gene with roles in neuronal wiring and extensive alternative splicing.

DOI - Digital Object Identifier

A persistent identifier for a publication.

PMID - PubMed Identifier

The identifier of a PubMed record.

SVG - Scalable Vector Graphics

An image format that preserves editable text and vector shapes.

PNG - Portable Network Graphics

A raster image format.

PDF - Portable Document Format

A document format suited to saving and printing the report.

CSV - Comma-Separated Values

A plain-text table format.

JSON - JavaScript Object Notation

A structured format used to save data and analysis results.

R - R programming language

The language used in the downloadable analysis scripts.

RDS - R serialized data

An R file containing a saved object.

HDF5 - Hierarchical Data Format, version 5

A format used to store the fitted MOFA model.

NA - Not Available

A missing value.

UBC - University of British Columbia

The institution offering BIOC557.

BIOC557 - Biochemistry course 557

The course associated with this lab.